Diveesha Vanchireddy

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Education

University of Wisconsin–Madison Madison, WI

Bachelor of Science in Data Science; Certificate in Economic Analytics *Expected May 2027*

- **Relevant Coursework:** Applied Econometrics, Statistical Modeling, Intro to Artificial Intelligence, Data Science Programming I & II

Professional Experience

AI Technical Lead Feb. 2026 – Present

Halverson Lab, UW–Madison (WISCURDS)

- Take **ownership** of the ChatECL AI platform for an education research lab, gathering requirements and shipping features (transcript editing, speaker labeling) via **CI/CD**.
- **Innovated** an AI-powered transcription workflow on Google Gemini after **evaluating models** on quality, cutting turnaround from **40 minutes to 2–3 minutes** per interview (over 90% faster).
- Turned ∼600 interview documents into a **PostgreSQL data model** with **Python, SQL, and LLMs**; built a read-only SQL tool and help maintain the **RAG** pipeline (Pinecone).
- **Leading a new district-comparison dashboard (in progress):** designed a staged **Google Cloud** pipeline (Cloud Storage → VM → clean table → separate Cloud SQL database) kept separate from production, with immutable raw files and plans for suppressed values.
- Built the platform's **cloud cost model** from billing data, verifying vendor rates against invoices, separating one-time from recurring costs, and deferring ∼ $30/month in database spend until data size was measured.
- Set up **least-privilege, upload-only** Cloud Storage access for teammates and coordinated distributed data uploads with a standard folder structure and privacy checks.

Lead Data Analyst Oct. 2024 – Present

Biokind Analytics, UW–Madison Pro Bono Consulting

- **Lead a team of 4–5 analysts** across client engagements for Dane County nonprofits (Diaper Village, United Way, Boys & Girls Club), from scoping through final presentation.
- Built **regression models** and **segmented** donor data, estimating per-donation value at $**200 (individual) vs.** $**2,884 (company)** to set suggested-donation amounts; **donor retention improved** after the client adopted our recommendations.
- Defined **KPIs** from scratch and presented **client recommendations** via Tableau and PowerPoint.

Undergraduate Research Assistant Feb. 2025 – Jul. 2026

Applied Population Lab, UW–Madison

- Designed and authored **ACSEST**, an automated R pipeline pulling Census Bureau API data across 8 geography levels with margins of error, caching, and unit tests; replaced a broken workflow for the team.
- Built **econometric regression models** across **3,133 U.S. counties** linking job market churn to health outcomes, controlling for poverty and unemployment and validating with **bootstrap resampling**.
- Integrated **multi-source data** (Census, BDS, County Health Rankings) with data quality checks.

Leadership

Marketing Chair & Founding Member Sep. 2026 – Present

Delta Phi Omega Sorority, Inc., UW–Madison Chapter

- Helped found the chapter; planned the **recruitment event** and reached **55K+ views in the first month**.

Projects

UrbanPlot | Business Viability Prediction Engine Feb. 2026

MadData Hackathon (24 hours) · Python, NLP, JavaScript

- Built a **predictive scoring model** ranking 676 vacant Madison commercial lots on business viability, using **NLP** on 7,466 web-scraped posts to measure demand.

CareCompass | AI Medical Triage Assistant (GitHub) Nov. 2025

ClaudeHacks (Anthropic-sponsored) · Honorable Mention · sentence-transformers, FAISS, Flask, React

- With a team, built a triage tool in 60 minutes using 384-dim embeddings and a **FAISS** index over 26 scenarios, returning urgency scores and UW Health facilities in **under 100ms**.

A/B-Tested Web Application Summer 2025

CS 320, UW–Madison · Python, HTML, CSS, JavaScript

- Ran an **A/B test** on button design, tracking click-through rate to make a **data-driven decision**.

Technical Skills

Programming & Data: Python (pandas, scikit-learn, PyTorch), SQL, R, JavaScript; PostgreSQL, Snowflake, BigQuery, ETL, Spark
Cloud & AI: Google Cloud (Cloud Storage, Cloud SQL, Vertex AI, IAM), LLMs, RAG, AI agents, vector search, Claude Code, Git, CI/CD
Analytics: regression, econometrics, hypothesis testing, A/B testing, NLP, cost analysis; Tableau, Power BI, Excel, Canva